Name_____ Date_____ Period_____

LESSON 8-1 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Equations and Solutions

Maryland Standard 6.EE.B.5 · Learning target – I can check whether a value is a solution of an equation by substituting it.

REMEMBER

- A **solution** is a value of the variable that makes an equation **true**.
- To check a value, **substitute** it for the variable. Then compare the two sides.
- If the two sides are not equal, the value is **not** a solution.

WORKED EXAMPLE

Is $n = 14$ a solution of $n + 9 = 23$?

Step 1 Replace n with 14: $14 + 9 = 23$

Step 2 Compare: $23 = 23$. Both sides match.

Answer **Yes, $n = 14$ is a solution.**

WORD BANK

equation – two amounts are equal

solution – the number that makes it true

substitute – put the number in for the letter

4p – 4 times p

PRACTICE

show your work · label units

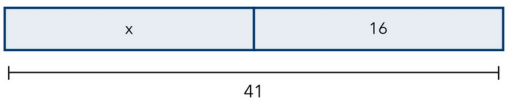
1. Substitute the value. Write **T** for true or **F** for false.

a. $y + 8 = 20$ $(y = 12)$ ____ **b.** $5m = 35$ $(m = 6)$ ____ **c.** $k - 9 = 16$ $(k = 25)$ ____

d. $h \div 4 = 7$ $(h = 32)$ ____ **e.** $13 + w = 30$ $(w = 17)$ ____ **f.** $6p = 54$ $(p = 9)$ ____

g. $q - 11 = 4$ $(q = 15)$ ____ **h.** $9r = 63$ $(r = 8)$ ____

2. The tape diagram shows $x + 16 = 41$. Which value is the solution: 15, 25, or 35? Substitute each value to check.



Solution_____

3. Bike Rack. A bike rack held b bikes. 7 bikes **left**. 18 bikes were still there. Kendra writes $b - 7 = 18$. Is $b = 11$ a solution? Is $b = 25$ a solution? Show both checks.

Show your work.

Solution_____

4. Find the Error. Andre says $p = 8$ is a solution of $4p = 12$ because $4 + 8 = 12$. Explain his mistake. Then find the real solution.

Frame *Andre is wrong because 4p means ___. The solution is p = ___.*

_____

_____

5. Stretch. Malik paid $42 for t movie tickets that cost $6 each, so $6t = 42$. Is $t = 36$ a solution? Is $t = 7$? Explain which value makes sense.

Step 1 Put 36 in for t. Does 6×36 equal 42?

Step 2 Put 7 in for t. Check again.

Solution_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Make up an equation for someone at home. Check their guess together by substituting.

Name_____ Date_____ Period_____

LESSON 8-1 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Writing Equations

Maryland Standard 6.EE.B.6 · Learning target – I can write an equation for a situation and test a proposed answer.

REMEMBER

- Use a **variable** for the unknown amount.
- "7 less than *n*" is $n - 7$, **not** $7 - n$.
- **Test** an answer by substituting it.

WORKED EXAMPLE

b books plus 15 more is 52 books. Test $b = 37$.

Step 1 Equation: $b + 15 = 52$

Step 2 Test: $37 + 15 = 52$, so it checks.

Answer $b + 15 = 52$; $b = 37$

WORD BANK

increased by – add (+)
less than – subtract (−)
times – multiply (×)
divided by – divide (÷)

PRACTICE

show your work · label units

1. Write an equation for each sentence. Use *n* for the number.

Step 1 Find the key word.

Step 2 Write *n*, the sign, the number, = and the total.

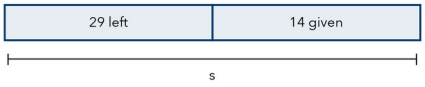
a. A number increased by 9 is 31. _____

b. 7 times a number is 63. _____

c. 12 less than a number is 20. _____

d. A number divided by 5 is 8. _____

2. Stickers. Rosa gave away 14 stickers and has 29 left. Write an equation for *s*. Test $s = 43$.



Equation and test_____

3. Field Trip. 56 students ride in 4 vans. Each van has the **same** number. Write an equation for *v*, the students in each van. Test $v = 12$ and $v = 14$.

Step 1 Vans × students in each van = all students.

Step 2 Put 12 in for *v*. Then try 14.

Show your work.

Equation and solution_____

4. Reading Log. Luis reads the same number of pages, p, each day for 5 days. He reads 85 pages in all. Write an equation. Test $p = 17$.

Step 1 Pages each day × 5 days = 85 pages.

Step 2 Put 17 in for p and check.

Equation and test_____

5. Find the Error. Tia is 13. She is 4 years older than her brother. Omar writes $b - 4 = 13$ for his age. Explain the mistake. Find b.

Frame *Omar is wrong because the brother is ___ than Tia. The equation should be ___, so b = ___.*

_____

_____

TEST PRACTICE

choose the best answer

6. Jada saved \$9 a week for w weeks. She saved \$72. Which equation fits?

A $w + 9 = 72$ **B** $9w = 72$ **C** $w \div 9 = 72$ **D** $72w = 9$

7. Which value of n is the solution of $n - 15 = 27$?

A $n = 12$ **B** $n = 27$ **C** $n = 42$ **D** $n = 405$

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___